

Agentic AI Debate Coach and Presentation Analysis Platform

PROJECT STATUS

Milestone 1 completed: core setup, authentication, profiles, role-based access control, and debate session management.

TECHNOLOGY

Frontend: React 19, Vite, React Router, Axios

Backend: Python, FastAPI, SQLAlchemy 2.0

Database: PostgreSQL with SQLite local fallback

Security: JWT access tokens and password hashing

Verification: Backend tests pass; frontend build passes

PRESENTATION SUMMARY

The first milestone establishes a working foundation for an AI-powered debate coaching platform. Users can create accounts, authenticate securely, manage profiles, schedule debate sessions, save transcripts, and receive coach evaluations.

SCOPE NOTE

Argument analysis, fallacy detection, speech analytics, AI debate simulation, and personalized AI coaching are planned for later milestones.

1. Project Objective and Scope

The platform helps users improve debating, public speaking, critical thinking, and persuasive communication. Milestone 1 focuses on the core application infrastructure and complete user-to-session workflow.

MILESTONE 1 OBJECTIVES

- Initialize the frontend and backend application structure.
- Create a secure authentication and authorization foundation.
- Support learner, coach, educator, and administrator roles.
- Create profile and skill-tracking fields for future coaching intelligence.
- Implement the debate session lifecycle from creation to evaluation.
- Document architecture, database entities, workflows, and tests.

COMPLETED AREAS

Authentication: register, login, logout, JWT persistence, protected routes

Profile: identity, bio, experience, topics, domains, goals, preferences

Dashboard: role-aware metrics, recent sessions, quick actions, skills

Sessions: create, schedule, filter, inspect, update, cancel, evaluate

Coach workflow: view learners, review sessions, submit scores and feedback

2. System Architecture

The application follows a layered full-stack architecture. React handles navigation, forms, dashboards, and user interactions. FastAPI exposes REST endpoints. SQLAlchemy provides persistence, while JWT dependencies protect authenticated operations.

ARCHITECTURE LAYERS

Client React 19 + Vite; pages, navigation, forms, dashboards
API FastAPI; validation, routing, and business rules
Security JWT and bcrypt/passlib; token and password protection
Persistence SQLAlchemy async; users, sessions, scores, feedback
Storage PostgreSQL in Docker; SQLite fallback for local development

BACKEND STARTUP

- FastAPI starts through backend/main.py.
- Database tables are created during development startup.
- CORS allows the Vite frontend to communicate with the API.
- Authentication, users, and sessions are mounted under the API prefix.
- PostgreSQL failure triggers fallback to debate.db.

3. Authentication and RBAC

Registration creates a user record with a securely hashed password and initialized coaching defaults. The backend returns a JWT containing the authenticated user identity and role.

AUTHENTICATION WORKFLOW

1. User submits email and password.
2. API verifies the stored password hash.
3. JWT access token is generated and returned.
4. Client stores the token and requests the profile.
5. Axios attaches the token to future API requests.
6. ProtectedRoute prevents unauthenticated page access.

ROLE PERMISSIONS

Learner	Own profile, own sessions, personal session creation
Coach	Learner permissions plus student roster and evaluations
Educator	Learner permissions plus student roster and evaluations
Admin	Broadest access, including system-level session deletion

Permissions are enforced in backend dependencies and route handlers. Frontend visibility improves usability but is not used as the security boundary.

4. Data Model and Profile Management

The database currently contains two main entities: User and DebateSession. A user owns many debate sessions. JSON columns provide flexible storage for profile collections and future machine-generated metrics.

USER ENTITY

Identity UUID, name, email, password hash, active status

Role learner, coach, educator, or admin

Preferences topics, presentation domains, goals, coaching preferences

Skills clarity, evidence strength, logic, persuasiveness, pace, confidence

Progress total sessions, average score, win streak, presentation history

Audit fields created and updated timestamps

PROFILE WORKFLOW

- User selects Beginner, Intermediate, Advanced, or Expert.
- Topics, domains, goals, and coaching preferences are multi-select tags.
- Changes are sent to PUT /users/me.
- Updated user data refreshes React state and local storage.
- Skill metrics display as progress indicators for future analytics.

5. Debate Session Workflow

Debate sessions are the central Milestone 1 workflow. A learner can prepare a session, schedule it, store speech content, and receive a formal evaluation from an authorized coach, educator, or administrator.

SESSION LIFECYCLE

Create Topic, format, position, date, duration, rounds, and notes
Scheduled New session is stored with scheduled status
Prepare User saves a transcript or argument notes
Review Privileged role views the session and transcript
Evaluate Evaluator submits a 0-100 score and feedback
Complete Session is completed and learner average is recalculated

SUPPORTED FORMATS

One-on-One Debate | Parliamentary Debate | Oxford Debate
Policy Debate | Public Forum Debate | AI Debate Simulation placeholder

SESSION API

GET	/sessions/	List sessions
POST	/sessions/	Create a scheduled session
GET	/sessions/{id}	View a session with access validation
PUT	/sessions/{id}	Update transcript or session fields
PUT	/sessions/{id}/evaluate	Submit evaluation and update average
DELETE	/sessions/{id}	Delete an authorized session

6. Frontend Experience and UI

The frontend is organized as a public landing/authentication area and a protected workspace. The visual pass preserves functionality while making the interface calmer and more consistent for an internship demonstration.

IMPLEMENTED SCREENS

- Landing page explaining platform direction and future capabilities.
- Login page with credential authentication and OAuth placeholder.
- Two-step signup with role selection and password strength feedback.
- Role-aware dashboard with metrics, sessions, quick actions, and skills.
- Debate Sessions page with filters, modals, transcripts, and evaluation.
- Profile page with editable identity data, skills, preferences, and goals.

VISUAL DESIGN CHANGES

- Replaced dark purple neon theme with a light neutral canvas.
- Added restrained teal accents and clearer typography.
- Replaced strong glows and oversized rounded surfaces with subtle borders.
- Added responsive layout behavior for smaller screens.
- Kept buttons, inputs, cards, modals, and badges consistent.

Development fallback: local sample sessions keep the interface demonstrable when no records exist or the API is unavailable.

7. Testing and Roadmap

The repository contains a Milestone 1 verification suite in `backend/tests/test_milestone1.py`. It exercises complete workflows rather than only isolated functions.

VERIFIED SCENARIOS

- Health and root endpoints respond successfully.
- All four roles can register.
- Valid login returns a token; invalid credentials return HTTP 401.
- Authenticated users can retrieve and update profiles.
- Learners can create sessions and save transcripts.
- Coaches can list students and view all sessions.
- Coaches can evaluate sessions and update learner averages.
- Unauthorized learners cannot evaluate or delete another session.
- Owners can delete their own sessions.

VALIDATION RESULTS

Backend Milestone 1 unittest suite: PASSED (OK)

Frontend production build: PASSED (Vite build completed)

Frontend lint: existing AuthContext warnings remain

ROADMAP

Milestone 1 Initialization, authentication, profiles, sessions - COMPLETED

Milestone 2 Argument analysis and fallacy detection - NEXT

Milestone 3 AI debate simulation and personalized coaching - PLANNED

Milestone 4 Presentation analytics, reports, deployment - PLANNED

8. Presentation Conclusion

For the first milestone, I built the working foundation of the DebateCoach AI platform. The system now supports secure role-based accounts, editable learner profiles, debate session scheduling, transcript storage, and coach evaluation.

I also documented the architecture and verified the main workflows with automated backend tests. The next phase will connect these stable workflows to AI services for argument analysis, fallacy detection, rebuttal generation, and adaptive debate coaching.

KEY TAKEAWAY

Milestone 1 converts the project proposal into a working, testable full-stack foundation. The data model and protected workflows are ready for intelligent analysis modules in the next milestones.

CURRENTLY NOT IMPLEMENTED

LLM extraction, fallacy detection, rebuttal generation, speech recognition, presentation analytics, real-time AI debate simulation, notifications, exports, and production monitoring are planned future work.

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